Learning Objectives 9.2–9.4: I can articulate what a Uniform Plane Wave (UPW) is and describe key properties that relate the electric and magnetic field. I can calculate the electric and magnetic fields for a UPW. I can articulate the concept of polarization and determine the polarization for various UPWs.

Show your work. Use the IEEE convention for handedness. Write a **Documentation** line at the end (**None** if you did not collaborate).

1. **A plane wave in a dielectric.** A uniform plane wave travels along $+z$ in a lossless dielectric with $\epsilon_r = 4$ and $\mu = \mu_0$. Its electric field is

$$\mathbf{E} = \hat{\mathbf{x}}\, 6\cos\left(2\pi \times 10^8\, t - \beta z\right) \text{ V/m}$$

(a) Starting from Faraday's law in phasor form, $\nabla \times \tilde{\mathbf{E}} = -j\omega\mu\tilde{\mathbf{H}}$, derive $\tilde{\mathbf{H}}$ for $\tilde{\mathbf{E}} = \hat{\mathbf{x}}\, E_0 e^{-j\beta z}$, and show that $E_0/H_0 = \sqrt{\mu/\epsilon}$.

Only the $\hat{\mathbf{y}}$ term of the curl survives, since $\tilde{\mathbf{E}}$ has only an x component and depends only on z:

$$\nabla \times \tilde{\mathbf{E}} = \hat{\mathbf{y}}\, \frac{d\tilde{E}_x}{dz} = -\hat{\mathbf{y}}\, j\beta E_0 e^{-j\beta z} = -j\omega\mu\tilde{\mathbf{H}}$$

$$\tilde{\mathbf{H}} = \hat{\mathbf{y}}\, \frac{\beta}{\omega\mu} E_0 e^{-j\beta z}$$

$$\frac{E_0}{H_0} = \frac{\omega\mu}{\beta} = \frac{\omega\mu}{\omega\sqrt{\mu\epsilon}} = \sqrt{\frac{\mu}{\epsilon}}$$

(b) Find β, λ, v_p, and η.

$$\beta = \frac{\omega\sqrt{\epsilon_r}}{c} = \frac{2\pi \times 10^8 \times 2}{3.00 \times 10^8} = 4.19 \text{ rad/m}$$

$$\lambda = \frac{2\pi}{\beta} = 1.50 \text{ m}, \qquad v_p = \frac{c}{\sqrt{\epsilon_r}} = 1.50 \times 10^8 \text{ m/s}$$

$$\eta = \frac{376.7}{2} = 188\ \Omega$$

β =	4.19 rad/m
λ =	1.50 m
v_p =	1.50×10^8 m/s
η =	$188\ \Omega$

(c) Write $\mathbf{H}(z,t)$.

$H_0 = 6/188.4 = 31.9$ mA/m, along $\hat{\mathbf{z}} \times \hat{\mathbf{x}} = \hat{\mathbf{y}}$:

$$\mathbf{H} = \hat{\mathbf{y}}\, 31.9\cos\left(2\pi \times 10^8\, t - 4.19z\right) \text{ mA/m}$$

2. **A wave along y.** In free space,

$$\mathbf{E} = \hat{\mathbf{z}}\, 3\cos\left(2\pi \times 10^9\, t - \beta y\right) \text{ V/m}$$

(a) Give the direction of travel and find β.

The phase $\omega t - \beta y$ is constant when y grows with t, so $\hat{\mathbf{k}} = \hat{\mathbf{y}}$.

$$\beta = \frac{\omega}{c} = \frac{2\pi \times 10^9}{3.00 \times 10^8} = 21.0 \text{ rad/m}$$

$\hat{\mathbf{k}}$ = $\hat{\mathbf{y}}$

β = 21.0 rad/m

(b) Find $\mathbf{H}(y, t)$, and check that $\mathbf{E} \times \mathbf{H}$ points along the direction of travel.

$$\tilde{\mathbf{H}} = \frac{1}{\eta_0}\hat{\mathbf{k}} \times \tilde{\mathbf{E}} = \frac{3}{376.7}\hat{\mathbf{y}} \times \hat{\mathbf{z}}\, e^{-j\beta y} = \hat{\mathbf{x}}\, 7.96 \times 10^{-3}\, e^{-j\beta y}$$

$$\mathbf{H} = \hat{\mathbf{x}}\, 7.96 \cos\left(2\pi \times 10^9\, t - 21.0 y\right) \text{ mA/m}$$

As a check, $\mathbf{E} \times \mathbf{H}$ points along $\hat{\mathbf{z}} \times \hat{\mathbf{x}} = \hat{\mathbf{y}}$, the direction of travel.

3. **Identify the polarization.** For each wave, state whether it is linear, circular, or elliptical, give the handedness of a circular or elliptical wave, and give the tilt from the x axis of a linear one.

(a) $\widetilde{\mathbf{E}} = (\hat{\mathbf{x}}\,3 + \hat{\mathbf{y}}\,4)\,e^{-j\beta z}$

Both components have the same phase, so $\Delta\phi = 0$ and the wave is linearly polarized, tilted at $\arctan(4/3) = 53.1^\circ$ from the x axis.

(b) $\widetilde{\mathbf{E}} = (\hat{\mathbf{x}} - \hat{\mathbf{y}})\,e^{-j\beta z}$

Since $-1 = e^{j180^\circ}$, $\Delta\phi = 180^\circ$, and the wave is linearly polarized along the line $y = -x$, at -45°.

(c) $\widetilde{\mathbf{E}} = 5\,(\hat{\mathbf{x}} + j\hat{\mathbf{y}})\,e^{-j\beta z}$

The amplitudes are equal and $\Delta\phi = +90^\circ$. The wave travels along $+z$ and $\sin\Delta\phi > 0$, so it is left-hand circularly polarized.

(d) $\mathbf{E} = \hat{\mathbf{x}}\cos(\omega t - \beta z) + \hat{\mathbf{y}}\,2\sin(\omega t - \beta z)$

Since $2\sin\theta = 2\cos(\theta - 90^\circ)$, $E_{x0} = 1$, $E_{y0} = 2$, and $\Delta\phi = -90^\circ$. The amplitudes differ, so the wave is elliptical, and $\sin\Delta\phi < 0$ along $+z$ makes it right-hand. At $z = 0$ the tip goes from $\hat{\mathbf{x}}$ at $t = 0$ to $\hat{\mathbf{y}}\,2$ at $T/4$, so the major axis is along y with half-length 2, the minor axis along x with half-length 1, and the axial ratio is 2.

(e) $\widetilde{\mathbf{E}} = (j\hat{\mathbf{x}} + \hat{\mathbf{y}})\,e^{+j\beta z}$

The wave travels along $-z$. The x component has phase $+90^\circ$ and the y component 0, so $\Delta\phi = \phi_y - \phi_x = -90^\circ$; with equal amplitudes, the wave is circularly polarized. At $z = 0$, $E_x = -\sin\omega t$ and $E_y = \cos\omega t$, so the tip goes from $\hat{\mathbf{y}}$ at $t = 0$ to $-\hat{\mathbf{x}}$ at $T/4$, counterclockwise about $+\hat{\mathbf{z}}$. With the right thumb along $-\hat{\mathbf{z}}$ the fingers curl the other way, so the wave is left-hand circularly polarized.

4. **The axes of an ellipse.** A wave along $+z$ has, at $z = 0$,

$$\mathbf{E} = \hat{\mathbf{x}} \cos \omega t + \hat{\mathbf{y}} \cos\left(\omega t + 60^\circ\right) \text{ V/m}$$

(a) Starting from $E_x^2 - 2E_xE_y \cos\Delta\phi + E_y^2 = \sin^2\Delta\phi$, find $|\mathbf{E}|$ where the path crosses the lines $y = x$ and $y = -x$.

On $y = x$, $E_y = E_x$:

$$E_x^2\left(2 - 2\cos\Delta\phi\right) = \sin^2\Delta\phi \quad \Longrightarrow \quad |\mathbf{E}|^2 = 2E_x^2 = \frac{\sin^2\Delta\phi}{1 - \cos\Delta\phi} = 1 + \cos\Delta\phi = 1.5$$

so $|\mathbf{E}| = 1.22$ V/m. On $y = -x$ the same steps give $|\mathbf{E}|^2 = 1 - \cos\Delta\phi = 0.5$, so $|\mathbf{E}| = 0.707$ V/m.

(b) Give the tilt of the major axis, the axial ratio, and the handedness.

The equation does not change when we swap E_x and E_y, so the axes lie on $y = \pm x$. The major axis is on $y = x$, at $+45^\circ$, and the axial ratio is $1.22/0.707 = 1.73$. Because $\sin 60^\circ > 0$ for a wave along $+z$, the wave is left-hand elliptically polarized.

tilt =	$+45^\circ$
axial ratio =	1.73
hand =	left

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